

Section 14: Cisco Router and Switch Basics

Layer: L2/L3 | Reference: Odom Vol1 Ch5-8: Implementing Ethernet LANs, Switch Interfaces | *Original study notes* — CCNA 200-301

Before configuring advanced features, you need fluency with the basic building blocks of Cisco routers and switches: interface naming, status verification, and the core difference in how each device forwards traffic.

Key Concepts

- A switch forwards frames based on destination MAC address using its MAC address table (CAM table), built by passively learning source MACs on incoming frames
- A router forwards packets based on destination IP address using its routing table, and is required to communicate between different subnets/VLANs
- Switch interfaces default to Layer 2 (switchport) mode; routers operate at Layer 3 by default on every interface
- Interface naming conventions: FastEthernet (100 Mbps), GigabitEthernet (1 Gbps), TenGigabitEthernet (10 Gbps) - always verify with 'show interfaces' rather than assuming from context
- Interface status has two key fields: line status (physical layer, Layer 1) and protocol status (Layer 2) - 'up/up' is healthy; 'down/down' usually means no cable or admin shutdown; 'up/down' suggests a Layer 2 mismatch (e.g., encapsulation)
- Auto-negotiation handles speed and duplex automatically on most modern Ethernet; mismatches (one side forced, other auto) cause collisions and performance issues
- Switches build their MAC table dynamically and age out entries after a period of inactivity (default 300 seconds on most platforms)
- A 'no switchport' command on a multilayer switch interface converts it from L2 to a routed L3 port

Command Reference

Command	Purpose
<code>show ip interface brief</code>	Summary of every interface's IP address and up/down status
<code>show interfaces <intf></code>	Detailed Layer 1/2 stats: line/protocol status, speed, duplex, errors
<code>show interfaces status</code>	Switch-specific: compact view of port status, VLAN, duplex, speed
<code>show mac address-table</code>	View the switch's learned MAC-to-port mappings
<code>speed {10 100 1000 auto}</code>	Manually set or auto-negotiate interface speed
<code>duplex {half full auto}</code>	Manually set or auto-negotiate interface duplex
<code>no switchport</code>	Convert a multilayer switch port from Layer 2 to a routed Layer 3 port

Configuration Walkthrough

Diagnose and fix a speed/duplex mismatch

```
SW1#show interfaces FastEthernet0/1
! Look for: many collisions/late collisions = duplex mismatch
SW1(config)#interface FastEthernet0/1
SW1(config-if)#speed 100
SW1(config-if)#duplex full
```

Convert a switch port to a routed port on a multilayer switch

```
SW1(config)#interface GigabitEthernet0/1
SW1(config-if)#no switchport
SW1(config-if)#ip address 10.10.10.1 255.255.255.0
```

Worked Scenario

Q: An interface shows 'up/down' in 'show ip interface brief'. What does that combination usually mean and what should you check first?

Line status up means Layer 1 (the physical cable/signal) is fine, but protocol status down points to a Layer 2 problem. Common causes include an encapsulation mismatch (e.g., HDLC vs PPP on serial links), a missing 'no shutdown' on the peer side combined with partial negotiation, or - on some platforms - a keepalive mismatch. Check 'show interfaces' for encapsulation type and keepalive settings on both ends.

Common Mistakes / Gotchas

- Assuming an interface name's speed from its label alone instead of verifying actual negotiated speed/duplex
- Mixing auto-negotiation on one side with a hard-coded speed/duplex on the other, causing a mismatch
- Forgetting MAC address table entries age out over time and will need to be relearned if a host has been quiet

Exam Tips

★ Memorize the line/protocol status combinations and what each implies (down/down, up/down, up/up, administratively down)

★ Know that switches forward on MAC (L2) while routers forward on IP (L3) - this distinction underlies nearly every other topic